Pediatric Hypoglycemia / Diabetic Emergency

PM-02 | Pediatric Medical

Quick flow

01 Check Blood Glucose [EMT / AEMT / Paramedic]

Assess airway, mental status, medications/insulin, intake, diabetes history, perfusion, and possible toxin exposure. Confirm unexpected CGM readings with capillary glucose when feasible.

02 Glucose <70 mg/dL? [EMT / AEMT / Paramedic]

Yes: treat hypoglycemia. Do not delay treatment of a symptomatic child for CGM confirmation. No: evaluate other causes and hyperglycemia/DKA features.

03 Can Swallow Safely? [EMT / AEMT / Paramedic]

Yes: oral glucose/food per local weight-based product. No: D10 IV/IO or IM glucagon.

04 D10 for Pediatric Hypoglycemia [AEMT / Paramedic]

D10 2 mL/kg IV/IO (0.2 g/kg); recheck glucose and neurologic status after 5 minutes; repeat once if hypoglycemia persists.

05 Suspected DKA + Dehydration? [AEMT / Paramedic]

Treat clinical dehydration/poor perfusion, not hyperglycemia alone: NS 10 mL/kg IV/IO, reassess, and repeat once only if needed; maximum 20 mL/kg before Medical Control.

06 Recheck + Transport [EMT / AEMT / Paramedic]

Repeat glucose and neurologic assessment. Monitor for DKA cerebral-edema signs; notify destination early and transport.

Overview

- Check blood glucose early in any child with altered mental status, seizure, weakness, or suspected diabetic emergency.
- Treat hypoglycemia promptly while protecting the airway. Evaluate hyperglycemia for dehydration, poor perfusion, and possible DKA without treating a glucose number alone.

Provider-Level Actions

Use a length- or weight-based pediatric system and reassess after every intervention.

EMT

- Check glucose and give oral glucose only if the child is awake, follows commands, and can swallow/protect the airway.
- If oral treatment is unsafe, support airway/ventilation and request ALS promptly. Do not delay treatment of symptomatic hypoglycemia for a confirmatory CGM or capillary reading.
- Use AO-02 Newly Born for immediate newborn resuscitation. After immediate stabilization, use this pathway for the approved pediatric hypoglycemia treatment.

AEMT

- Perform all EMT care plus IV/IO access and D10 2 mL/kg (0.2 g/kg); recheck after 5 minutes and repeat once for persistent hypoglycemia.
- Use glucagon when vascular access is unavailable and oral therapy is unsafe. For suspected DKA with dehydration/poor perfusion, give cautious normal saline as directed and reassess.

Paramedic

- Perform all prior care plus cardiac monitoring, evaluation for alternate causes, and management of dysrhythmia, seizure, shock, or airway compromise.
- Do not give insulin or sodium bicarbonate in the field for suspected pediatric DKA. Contact Medical Control early for worsening neurologic status, shock, or suspected cerebral edema.

Indications

- Pediatric patient with glucose below 70 mg/dL, significant hyperglycemia, or symptoms suggesting a diabetic emergency.

Treatment steps

- Treat glucose below 70 mg/dL when symptomatic or when symptom assessment is unreliable. Recheck glucose and neurologic status after every intervention.
- If a CGM reading is unexpectedly low or high, confirm with capillary glucose when feasible; do not delay treatment in the symptomatic child.
- Hyperglycemia alone does not diagnose DKA. Evaluate dehydration, perfusion, vomiting, abdominal pain, Kussmaul respirations, and mental-status change.
- For suspected DKA, do not give insulin, sodium bicarbonate, or routine/repeated large fluid boluses in the field.
- Watch for cerebral-edema warning signs: worsening headache, altered mental status, recurrent vomiting, bradycardia, hypertension, or abnormal respirations. Request rapid transport and Medical Control if present.
- Refusal after hypoglycemia requires normal neurologic status/capacity, glucose at least 80 mg/dL, ability to eat, a reliable caregiver, a clear reversible cause, and no long-acting oral hypoglycemic-agent concern.

Medications

Oral glucose

Per local product and weight-based system; give only when the child can swallow and protect the airway

Dextrose 10% (D10)

2 mL/kg IV/IO (0.2 g/kg); recheck glucose and neurologic status after 5 minutes and repeat once for persistent hypoglycemia. Contact Medical Control for additional dextrose dosing or infusion.

Glucagon

<20 kg: 0.5 mg IM; ≥20 kg: 1 mg IM when oral glucose is unsafe and vascular access cannot be obtained promptly

Normal saline

For suspected DKA with dehydration/poor perfusion: 10 mL/kg IV/IO, reassess, and repeat once only if needed; maximum 20 mL/kg before Medical Control

Warnings / precautions

- Do not administer oral glucose when swallowing or airway protection is impaired.
- Do not give insulin or sodium bicarbonate in the field for suspected pediatric DKA.
- Do not give large or repeated fluid boluses routinely in suspected pediatric DKA.
- Do not delay treatment of symptomatic hypoglycemia for a confirmatory glucose reading.

Clinical pearls

- D10 2 mL/kg provides 0.2 g/kg of dextrose and is the approved pediatric IV/IO hypoglycemia dose.
- Pediatric DKA can deteriorate neurologically; avoid over-resuscitation and communicate serial mental-status findings to the receiving team.

Related pathways / contacts

AO-02 Newly Born - Open for immediate newborn resuscitation before transitioning to the approved hypoglycemia pathway.

AR-05 Pediatric Airway - Open when hypoglycemia causes unsafe airway protection or ventilation.

PM-03 Pediatric Hypotension / Shock - Open for persistent poor perfusion or shock.

Medication Reference - Open the current Claiborne medication reference.

Source and references

Claiborne County EMS PM-02 Pediatric Hypoglycemia / Diabetic Emergency protocol, approved October 26, 2026. | ISPAD Clinical Practice Consensus Guidelines: Diabetic Ketoacidosis and Hyperglycemic Hyperosmolar State. | American Diabetes Association Standards of Care: Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises.

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